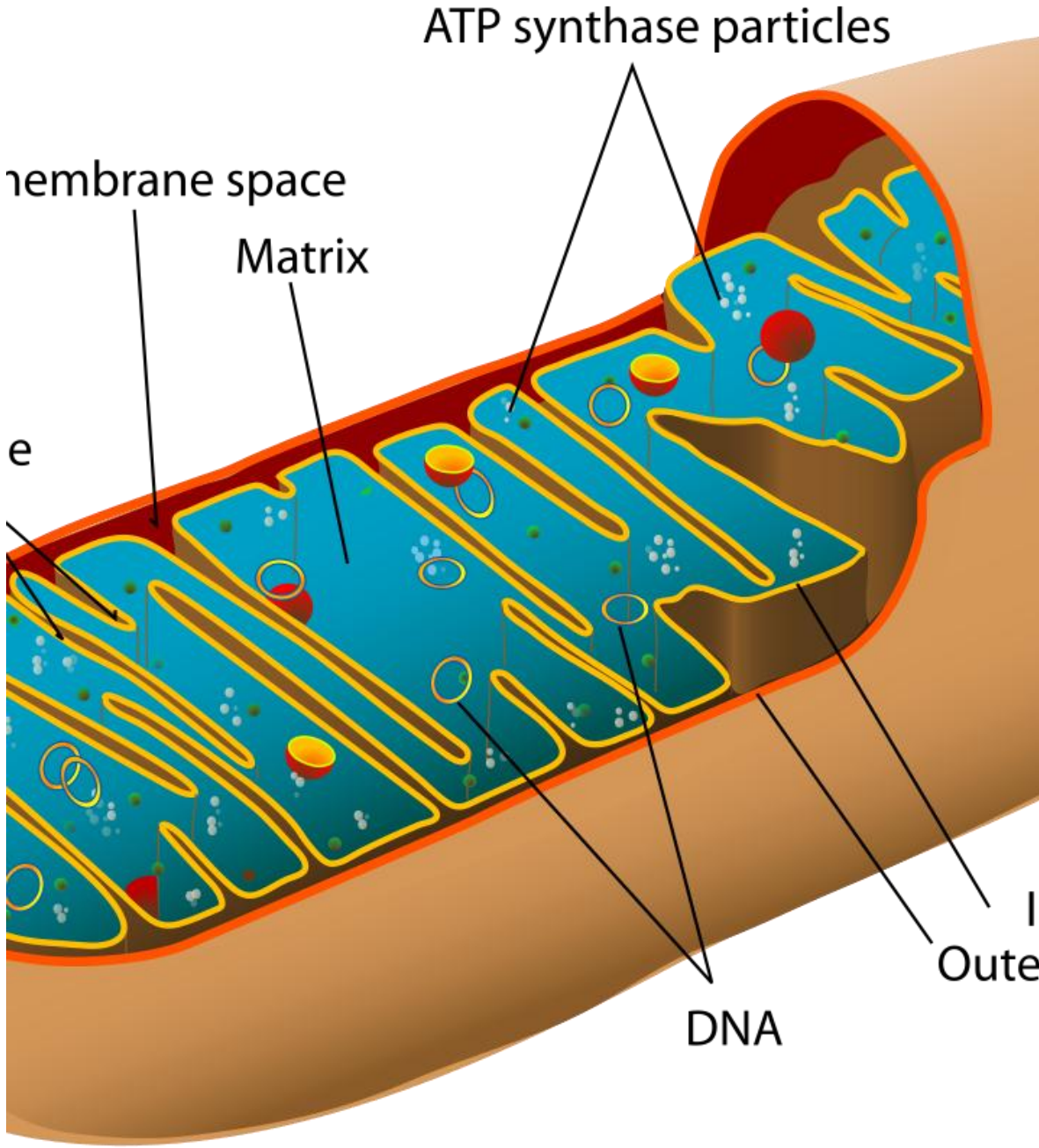
DP Biology – C1.2 (HL extension)

Inside cell respiration

- What do hydrogen and oxygen contribute to releasing energy in cells?
- How is energy shared out and used within cells?



HL statements C1.2.7–17. Textbook pp. 412–424.
Image: a mitochondrion. Mariana Ruiz Villarreal, public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.2.7–8	Explain oxidation as removal of hydrogen, the role of NAD, and the stages of glycolysis
2	C1.2.9–10	Explain how anaerobic respiration regenerates NAD in humans and in yeast, and its uses in baking and brewing
3	C1.2.11–12	Explain the link reaction and the Krebs cycle, and tally their products
4	C1.2.13–14	Explain how reduced NAD passes energy to the electron transport chain, which builds a proton gradient
5	C1.2.15–16	Explain chemiosmosis, ATP synthase and the role of oxygen as the final electron acceptor
6	C1.2.17	Compare fats and carbohydrates as fuels for respiration

Oxidation, reduction and NAD

	Oxidation	Reduction
Electrons	Lost	Gained
Hydrogen	Removed	Added
Oxygen	Added	Removed

Respiration **oxidizes** glucose by removing hydrogen (**dehydrogenation**).

The coenzyme **NAD** accepts the hydrogen and is **reduced**: $NAD^+ + 2H \rightarrow NADH + H^+$.

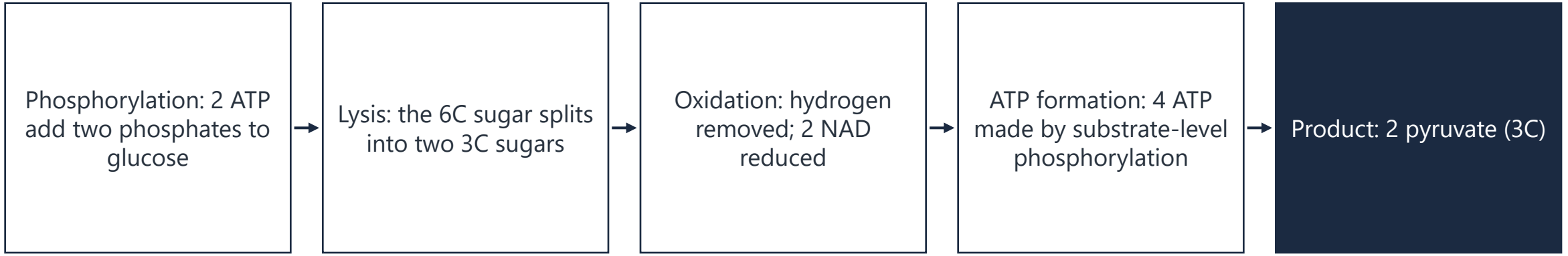
Oxidation and reduction always happen together: **redox**.

Key idea

A hydrogen atom is a proton plus an electron, so removing hydrogen removes electrons: oxidation. NAD carries the hydrogen, and its energy, to the electron transport chain.

Glycolysis: glucose to pyruvate

C1.2.8



Per glucose	Amount
ATP used	2
ATP made	4
Net ATP	2
Reduced NAD	2

Where and how

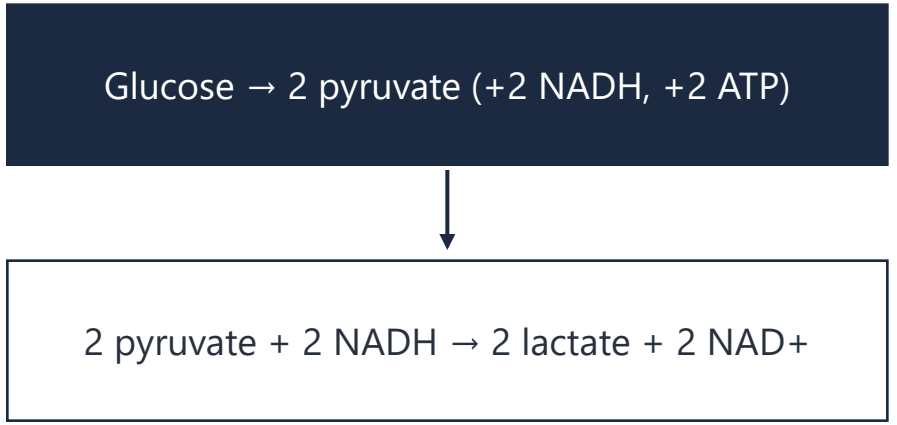
In the cytoplasm, with no oxygen needed. Each of the ten or so steps has its own enzyme. Phosphorylation makes glucose less stable, so it can be split.

Regenerating NAD without oxygen

C1.2.9

Glycolysis needs a constant supply of NAD^+. Without oxygen, reduced NAD cannot pass its hydrogen to the electron transport chain, so NAD^+ would run out and glycolysis would stop.

In humans, **pyruvate** accepts the hydrogen and becomes **lactate**. This regenerates NAD^+, so glycolysis can continue, giving a net 2 ATP per glucose.

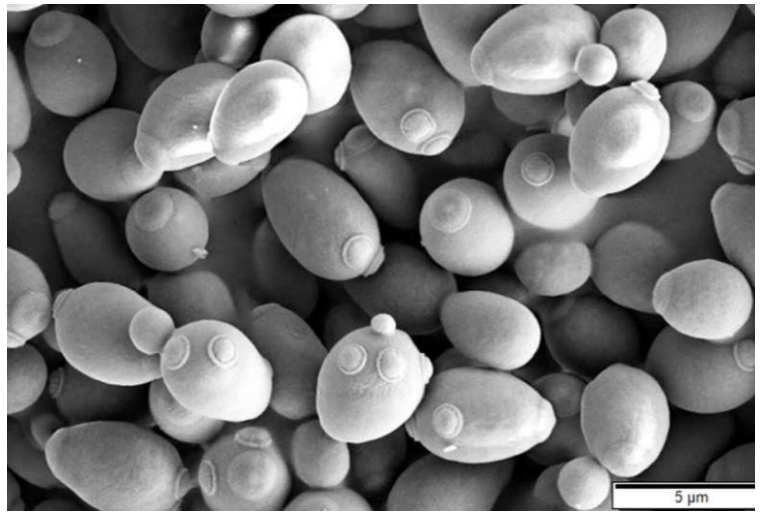


In muscle

During intense exercise, muscles cannot get oxygen fast enough and make lactate. This gives ATP quickly but little of it. Lactate is later taken to the liver and converted back, using oxygen.

Anaerobic respiration in yeast

C1.2.10



Yeast cells (SEM). M. D. Murtey and P. Ramasamy, CC BY 3.0.



Bread dough rising in baskets. Wald1siedel, CC BY-SA 4.0.

	Humans	Yeast
Same	Glycolysis	Glycolysis
Hydrogen acceptor	Pyruvate	Ethanal (from pyruvate)
Products	Lactate	Ethanol and carbon dioxide

Use	Product that matters
Baking	Carbon dioxide makes dough rise; ethanol evaporates in the oven
Brewing	Ethanol; carbon dioxide gives fizz

The link reaction

C1.2.11

Pyruvate (3C) enters the matrix → Decarboxylation: CO_2 removed → Oxidation: hydrogen removed, NAD reduced → Acetyl group (2C) joins coenzyme A → **Acetyl-CoA enters the Krebs cycle**

Other fuels

Fatty acids from lipids are cut into 2C acetyl groups, which also join coenzyme A. So acetyl-CoA is the common entry point to the Krebs cycle for both carbohydrates and lipids.

Per glucose (2 pyruvate)	Yield
CO_2	2
Reduced NAD	2

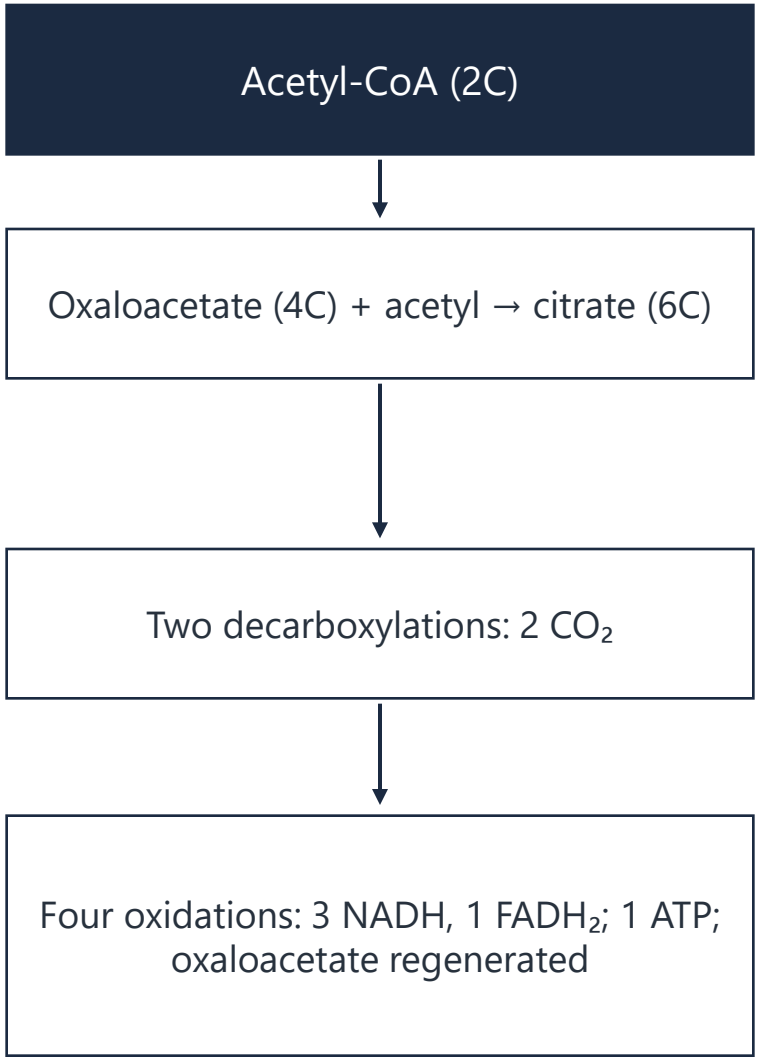
The Krebs cycle

The **Krebs cycle** takes place in the mitochondrial **matrix**.

Each turn adds a 2C acetyl group to 4C **oxaloacetate**, making 6C **citrate**.

Citrate is gradually oxidized and decarboxylated back to oxaloacetate.

It turns **twice** per glucose.



Per turn	Yield
CO_2	2
Reduced NAD	3
Reduced FAD	1
ATP	1

Where the hydrogen goes

C1.2.8–12

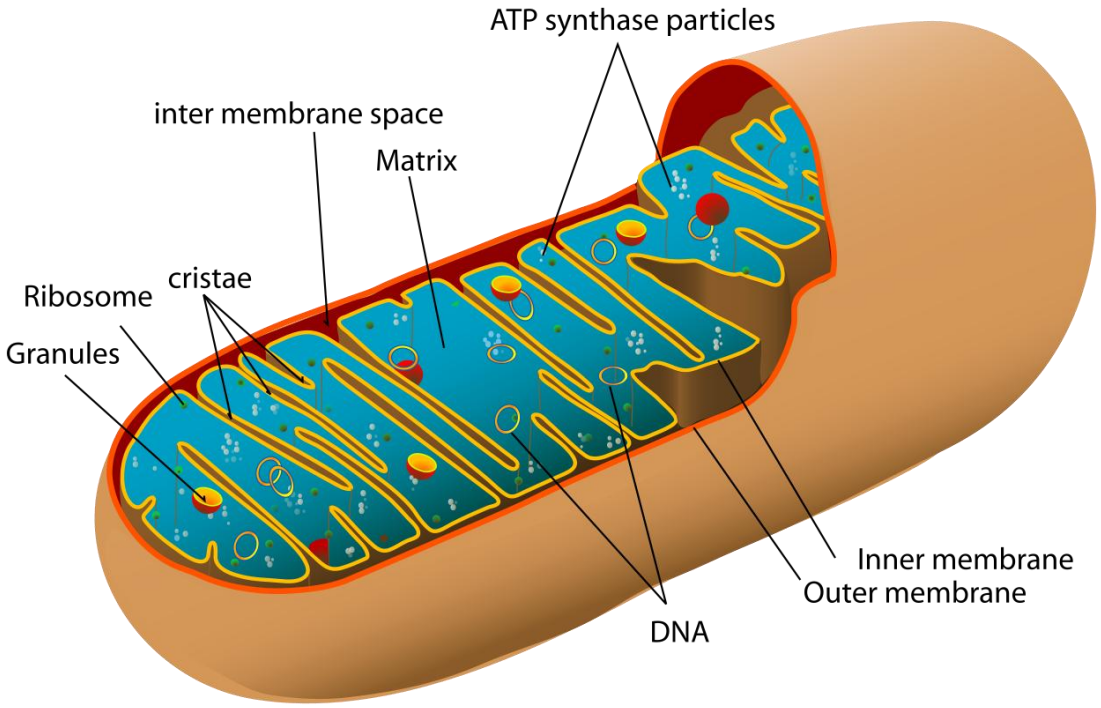
Stage, per glucose	ATP	Reduced NAD	Reduced FAD	CO_2
Glycolysis	2 (net)	2	0	0
Link reaction	0	2	0	2
Krebs cycle (two turns)	2	6	2	4
Total so far	4	10	2	6

Activity

Complete this table from memory, then explain why most of the energy of glucose is still not in ATP at this stage. Where is it?

Reduced NAD delivers electrons to the chain

C1.2.13



Mitochondrion: the chain is in the inner membrane (cristae). Public domain.

Reduced NAD from glycolysis, the link reaction and the Krebs cycle passes a pair of high-energy **electrons** to the first carrier of the **electron transport chain**. This oxidizes it back to NAD^+, ready to be reused.

Reduced FAD passes its electrons to a later carrier, so it releases less energy.

The carriers are proteins embedded in the **inner mitochondrial membrane**, folded into cristae to give a large surface area.

Building a proton gradient

C1.2.14

Electrons pass from carrier to carrier → Each step releases a little energy → **Energy pumps protons from the matrix into the intermembrane space** → A high proton concentration builds up in the small space

Why in small steps?

Releasing all the energy at once would waste it as heat. Passing electrons along a chain of carriers lets the energy be used bit by bit to pump protons. The inner membrane is impermeable to protons, so they can only return through ATP synthase.

Chemiosmosis and ATP synthase

C1.2.15

Protons diffuse back into the matrix **down their concentration gradient**. The only route is through **ATP synthase**.

The flow of protons turns part of the enzyme, changing its shape, and this energy joins ADP and phosphate to make ATP. This is **chemiosmosis**, and the whole process is **oxidative phosphorylation**.

In total, about 30 to 32 ATP are made per glucose.

Mitchell's idea

Peter Mitchell proposed chemiosmosis in 1961. Many biochemists rejected it at first. Experiments showing that a proton gradient alone could drive ATP synthesis convinced them, and he received a Nobel Prize in 1978.

Common mistake

Protons are not used up in making ATP. Their movement down the gradient provides the energy.

Oxygen: the final electron acceptor

At the end of the chain: 2 electrons + 2 H^+ (from the matrix) + 1/2 O_2 → H_2O

Oxygen accepts the electrons from the last carrier and combines with protons from the matrix, forming **metabolic water**.

Removing electrons at the end keeps them flowing along the chain. Without oxygen, the chain backs up, reduced NAD cannot be recycled, and only glycolysis can continue.

Cyanide

Cyanide blocks the last carrier, so electrons cannot reach oxygen. The chain stops, ATP production collapses, and cells die even though oxygen is present.

Lipids or carbohydrates?

C1.2.17

	Carbohydrate	Lipid
Energy per gram	About 17 kJ	About 37 kJ
Why	More oxygen already in the molecule	Less oxygen; more hydrogen to oxidize
Entry to respiration	Glycolysis, then link reaction	Fatty acids cut into acetyl groups, enter as acetyl-CoA
Anaerobic use	Yes, via glycolysis	No: fatty acids do not go through glycolysis

So

Lipids are a compact energy store, but they can only be respired aerobically. Sprinters rely on carbohydrate; long endurance events use more fat.

Check-in: following the hydrogen

C1.2.7–16

1. Explain why removing hydrogen from a molecule is oxidation.
2. State how many ATP and reduced NAD glycolysis gives per glucose, after costs.
3. Explain why pyruvate is converted to lactate in anaerobic respiration.
4. Describe what happens to pyruvate in the link reaction.
5. Explain how the flow of protons through ATP synthase produces ATP.
6. Explain why cyanide stops ATP production.

How to use this

Answer without your notes, then compare with a partner.

Working with data: yeast and sugar concentration

C1.2.10

Volume of carbon dioxide released by yeast in 20 minutes at 30 °C, with different glucose concentrations. These data are invented for practice; the pattern is realistic.

Glucose / %	CO_2 / cm^3
0	0.2
2	4.1
4	7.6
8	10.8
16	11.2
32	6.5

1. Describe the effect of glucose concentration on the volume of CO_2.
2. Explain the increase up to 8%.
3. Suggest why the volume levels off between 8 and 16%.
4. Suggest why the volume falls at 32% glucose.

Common mistakes to avoid

Not this	But this
Lactate formation makes extra ATP.	It makes no ATP; it regenerates NAD+ so glycolysis can continue.
Glycolysis happens in the mitochondrion.	Glycolysis is in the cytoplasm; the link reaction and Krebs cycle are in the matrix.
Oxygen is used in the Krebs cycle.	Oxygen is only used at the end of the electron transport chain.
Protons are used up to make ATP.	Protons flow through ATP synthase; their movement provides the energy.
Yeast makes lactate.	Yeast makes ethanol and carbon dioxide.

Check that you can...

1. **Explain** oxidation as removal of hydrogen, and NAD as a hydrogen carrier
2. **Outline** the four stages of glycolysis and its net yield
3. **Explain** how lactate and ethanol formation regenerate NAD, and uses of yeast
4. **Explain** the link reaction and the Krebs cycle
5. **Explain** how the electron transport chain builds a proton gradient
6. **Explain** chemiosmosis and the role of ATP synthase
7. **Explain** the role of oxygen as the final electron acceptor
8. **Compare** fats and carbohydrates as fuels for respiration

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. How many ATP and reduced NAD does glycolysis give, per glucose, after costs?
2. Explain why anaerobic respiration must convert pyruvate to lactate.
3. Explain the role of oxygen in aerobic respiration.

Image credits

Animal mitochondrion diagram en.svg – Mariana Ruiz Villarreal LadyofHats – Public domain

Saccharomyces cerevisiae SEM.jpg – Mogana Das Murtey and Patchamuthu Ramasamy – CC BY 3.0

Bread dough in baskets 02.JPG – Wald1siedel – CC BY-SA 4.0

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